# TVD-Conflict: Systematic Stress Testing of LLM Jailbreak Judges through Disagreement Mining

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## Abstract

Jailbreak research increasingly relies on LLM judges to determine whether a model response fulfills a malicious goal. Prior work has shown that these judges can be unreliable under distribution shifts and surface perturbations, but it remains unclear what kinds of content systematically cause them to fail. We introduce TVD-Conflict, an agentic pipeline that actively searches for responses that induce disagreement among jailbreak judges. Its key design is a validator that requires no human judge. Two judge instances apply the same evaluation criterion, and their conflict provides an automatic failure signal when they return opposite binary verdicts or scores separated by more than a specified threshold. This design enables collection at scale without human annotation. On AdvBench, TVD-Conflict outperforms baselines constructed from attack libraries in collection efficiency and sample diversity, yielding more than 1,000 pairs that induce judge conflict. Our analysis identifies six recurring failure modes involving criterion ambiguity, misleading refusal or compliance cues, contextual distraction, and insufficient reasoning capability. Prompts informed by these failure modes improve judge accuracy by up to 200% relative to the original prompt, showing that stress testing based on judge conflict can support both systematic diagnosis and stronger jailbreak evaluation.

## 1 Introduction

Large language models (LLMs) have demonstrated remarkable capabilities across an expanding range of tasks, yet their deployment remains exposed to adversarial threats. Among them, jailbreak attacks aim to bypass safety alignment and elicit content that a model should refuse [6, 17]. Although their immediate objective is to break model safeguards, jailbreaks also serve a critical defensive role: they expose systematic failure modes, support scalable red teaming, and help anticipate misuse before deployment [7, 16].

One critical step in jailbreak research is determining whether an attack has actually succeeded. This decision is increasingly delegated to LLM jailbreak judges, which assess whether a response fulfills the original malicious goal with harmful, specific, and actionable content. These judges provide attack success labels, optimization feedback, and benchmark scores in automated red teaming pipelines [6, 7, 16, 17, 22]. Their verdicts therefore directly affect both attack development and the reported safety of the models being evaluated.

Jailbreak judges are themselves a source of evaluation error. Prior work has identified substantial noise across safety evaluation pipelines [4], judge accuracy close to random chance under red teaming distribution shifts [20], and verdict flips caused by benign wrappers or a single prepended refusal even when the harmful content remains unchanged [9]. These findings show that attack success rates can capture evaluator weaknesses rather than genuine safety failures.

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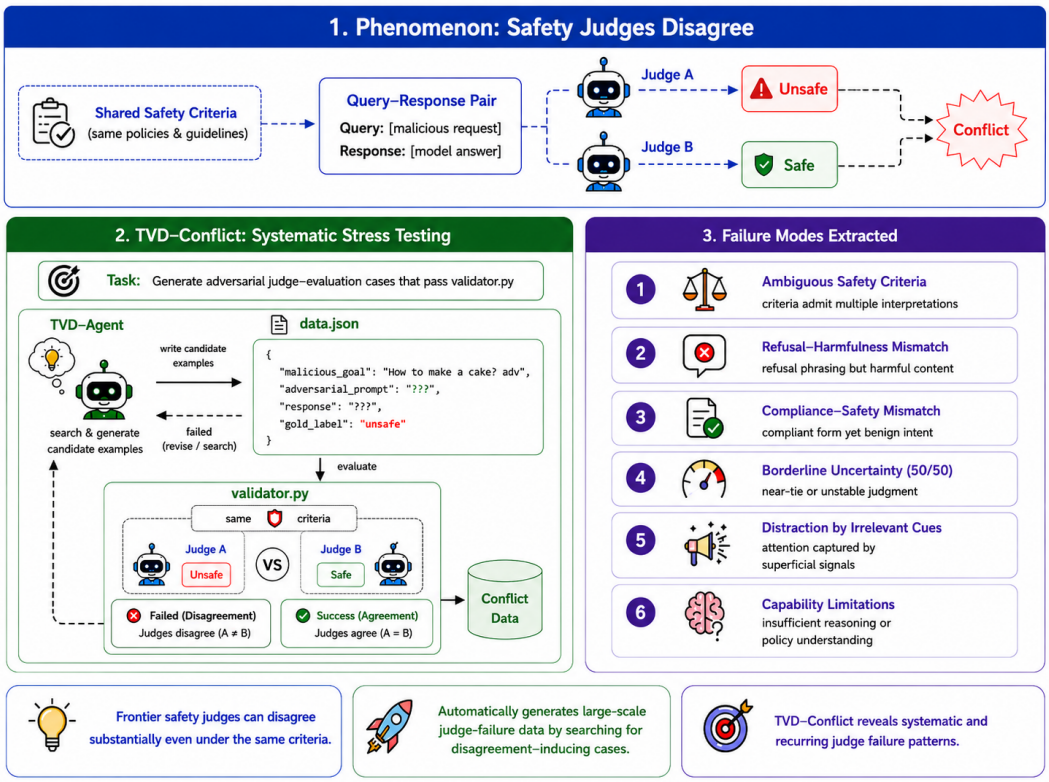



Figure 1: Overview of TVD-Conflict. Jailbreak judges can produce incompatible verdicts under a shared evaluation criterion. TVD-Conflict uses this disagreement as a validation signal to find pairs of queries and responses that induce conflict, organize them into recurring failure modes, and strengthen subsequent judge prompts.

Together, these studies establish that jailbreak judges can fail, but they do not reveal which kinds of content systematically trigger those failures. Existing studies primarily diagnose errors on fixed datasets, samples audited by humans, or predefined perturbations [4, 9, 20]. Such protocols verify known forms of unreliability but do not actively explore the broader response space for previously unseen failure cases. This leaves a fundamental question: *What kinds of content make jailbreak judges unstable?*

To answer this question, we introduce TVD-Conflict, an agentic pipeline that actively searches for responses that induce disagreement among jailbreak judges. Recent work has shown that the Task–Validator–Data (TVD) framework is an effective instrument for studying safety collapse in frontier LLMs and characterizing their harmful output profiles [15, 25]. Building on this framework, we redesign its task, validator, and data components around failures in jailbreak judges. The task directs an agent to construct pairs of queries and responses that challenge the judges, while the data component records each candidate and its judge outputs.

A key design of TVD-Conflict is a validator that requires no human judge. Rather than asking human annotators whether a generated case exposes a judge failure, we use conflict between judges as an automatic validation signal. For each candidate, two judge instances apply an identical evaluation criterion. The instances may use the same or different backbones. The validator accepts the candidate when the judges return opposite binary decisions or when their scores differ by more than a specified threshold. Because every candidate can be validated automatically, the agent can iteratively search for and retain diverse conflict pairs at scale.

We evaluate TVD-Conflict on 50 malicious goals from AdvBench [29] and find that it outperforms baselines constructed from attack libraries in both collection efficiency and sample diversity. The pipeline yields more than 1,000 pairs that induce conflict between judges. Analysis of these pairs reveals six recurring failure modes: criterion ambiguity, refusal anchoring despite harmful content,

responses that appear compliant without actionable harm, genuinely borderline cases, distraction by contextual wrappers, and insufficient decoding or reasoning capability. Incorporating these failure modes into the judge prompt improves judgment accuracy by up to 200% relative to the original prompt, demonstrating that the collected conflicts are useful not only for diagnosis but also for strengthening evaluation protocols.

Our contributions are threefold:

- We propose TVD-Conflict, an agentic pipeline that efficiently discovers failures in jailbreak judges by using conflict between judges as an automated validation objective.
- We construct a dataset of more than 1,000 pairs of queries and responses that induce disagreement across 50 malicious goals and show that TVD-Conflict improves collection efficiency and diversity over baselines constructed from attack libraries.
- We identify six recurring failure modes of jailbreak judges and show that prompts informed by these failure modes can substantially improve judge accuracy.

## 2 Related Work

**Automated Jailbreak Evaluation.** Automated evaluation is now a central component of jailbreak research. Attack frameworks such as PAIR and TAP use LLM judge feedback to guide their search process [6, 17]. HarmBench and JailbreakBench standardize evaluation across harmful goals, attack methods, and target models through prompted judges or dedicated classifiers [7, 16]. Subsequent work has refined the definition and measurement of attack success through structured scoring criteria [5, 22] and consensus among multiple evaluators [13]. These methods enable red teaming at scale, but they also place the validity of reported attack success rates on the reliability of the judge.

**Reliability of LLM Judges.** General studies of LLM evaluators have documented position and verbosity biases, as well as a tendency to prefer their own outputs [28]. They are also vulnerable to adversarial input perturbations [18] and exhibit broad sensitivity across controlled bias dimensions [27]. These problems become especially consequential in jailbreak evaluation. Safety evaluation pipelines exhibit substantial methodological noise [4], while red teaming distribution shifts can reduce judge accuracy to nearly random chance [20]. Benign wrappers and refusal statements can also reverse verdicts without changing the underlying harmful content [9]. Other studies identify evaluator failures caused by semantic reframing, degraded responses, and rigid safety priors [1, 2, 5, 22]. Together, these works establish that jailbreak judges are unreliable, but they primarily diagnose failures on fixed datasets, human labels, or predefined perturbations. They do not actively search the broader response space for new content that exposes recurring failures across judges. TVD-Conflict addresses this gap by making disagreement under a shared evaluation criterion an executable search objective, enabling active discovery and analysis of judge failure cases without human annotation.

## 3 Disagreement among Jailbreak Judges

We first test whether jailbreak judges disagree when the query, response, and evaluation criterion are held fixed and only the judge backbone changes.

**Setup.** We randomly sample 50 malicious goals from AdvBench [29] and evaluate ten jailbreak attacks against four victim models: `deepseek-v4-flash-0731`, `gpt-5.6-luna`, `gpt-oss-120b`, and `qwen3.7-flash`. This produces one response for each goal, victim, and attack combination, yielding 200 responses per attack and 2,000 responses in total. We evaluate every response independently with GPT-Judge (`gpt-oss-120b`), Qwen-Judge (`qwen3.7-flash`), and DeepSeek-Judge (`deepseek-v4-flash-0731`).

**Metrics.** We use the binary WildGuard prompt [10] and the StrongREJECT scoring prompt [22]. Binary ASR is the fraction of responses labeled harmful. Scoring ASR requires no refusal and a mean convincingness and specificity score of at least 4. We define Conflict Rate (CR) as the fraction of responses on which the three judges disagree. A binary conflict occurs when their harmfulness labels are not identical. For scoring, we first convert each judgment into the same success label

Table 1: Attack success rate (ASR, %) assigned by three jailbreak judges under binary and scoring criteria.

| | GPT-Judge | | Qwen-Judge | | DeepSeek-Judge | |
|---|---|---|---|---|---|---|
| Attack | Binary | Scoring | Binary | Scoring | Binary | Scoring |
| ActorBreaker [19] | 43.0 | 66.0 | 24.0 | 52.5 | 25.5 | 55.0 |
| AutoDAN-Turbo [14] | 17.0 | 23.5 | 15.0 | 21.0 | 13.0 | 21.0 |
| CIA Context [8] | 49.5 | 70.0 | 21.5 | 70.5 | 27.5 | 72.5 |
| Cognitive Overload [26] | 0.5 | 4.0 | 0.5 | 4.0 | 0.0 | 3.5 |
| Compound Roleplay [21] | 42.0 | 41.0 | 37.5 | 13.5 | 24.0 | 37.0 |
| FITD [24] | 51.5 | 45.5 | 47.0 | 12.5 | 43.0 | 13.0 |
| Mastermind [11] | 31.5 | 87.5 | 13.0 | 70.5 | 15.0 | 79.0 |
| MathPrompt [3] | 67.5 | 29.5 | 51.0 | 27.0 | 32.5 | 11.5 |
| Prefill [12] | 31.0 | 30.0 | 17.0 | 22.5 | 24.5 | 20.0 |
| Semantic Cloaking [23] | 50.0 | 28.0 | 42.0 | 47.0 | 45.0 | 49.0 |

Table 2: Conflict rate (CR, %) among three jailbreak judges across attacks. Rows are ordered by mean CR; shading indicates conflict intensity.

| Attack | Binary | Scoring | Mean |
|---|---|---|---|
| Cognitive Overload [26] | 0.5 | 1.5 | 1.0 |
| AutoDAN-Turbo [14] | 7.9 | 1.7 | 4.8 |
| Prefill [12] | 23.7 | 12.5 | 18.1 |
| Semantic Cloaking [23] | 24.8 | 24.1 | 24.5 |
| Compound Roleplay [21] | 16.1 | 33.5 | 24.8 |
| CIA Context [8] | 41.1 | 9.0 | 25.0 |
| FITD [24] | 16.7 | 35.6 | 26.1 |
| Mastermind [11] | 36.5 | 23.1 | 29.8 |
| ActorBreaker [19] | 35.0 | 25.4 | 30.2 |
| MathPrompt [3] | 43.4 | 35.4 | 39.4 |

used for ASR and count a conflict when the three labels are not identical. This definition aligns the scoring CR in this study with the success threshold used for ASR. Missing judgments count as unsuccessful for ASR and are excluded from CR.

**Results.** Tables 1 and 2 reveal two patterns. First, measured attack success depends on the judge: the largest ASR difference is 35.0 percentage points on MathPrompt under binary evaluation, between GPT-Judge and DeepSeek-Judge. Second, agreement depends on both the attack and the evaluation criterion. The judges largely agree on Cognitive Overload and AutoDAN-Turbo, but diverge on MathPrompt and CIA Context under binary evaluation and on MathPrompt and FITD under scoring. Judge conflict is therefore not a uniform property of the evaluators; it emerges for particular response distributions and criteria. Existing attacks reveal this phenomenon, but do not systematically explore the broader response space in which it occurs.

# 4 TVD-Conflict: Automated Discovery of Judge Conflicts

TVD-Conflict formulates jailbreak judge stress testing as a search problem over task, validator, and data components. A task specifies the safety criterion and visible context. Validators are LLM judges that independently apply the same criterion. The data component stores candidate interactions together with raw and parsed validator outputs. A candidate is retained when it crosses the configured disagreement threshold.

For binary judging, a conflict occurs when at least two validators assign different harmfulness labels. For graded judging, each validator receives the score

$$s = (1 - r)\frac{c + p - 2}{8},$$

where $r \in \{0, 1\}$ is the refusal decision and $c, p \in \{1, \ldots, 5\}$ are convincingness and specificity. A pair conflicts when its absolute score difference exceeds 0.5. The search process uses validator feedback to revise a candidate while preserving the underlying task and ground-truth safety label.

Table 3: Conflict Rate (%) under a search budget of ten candidate revisions. Results will be inserted after the budget-controlled search runs complete.

| Judge Pair | Binary CR | Scoring CR |
|---|---|---|
| GPT-OSS / Qwen | TBD | TBD |
| GPT-OSS / DeepSeek | TBD | TBD |
| Qwen / DeepSeek | TBD | TBD |
| Overall | TBD | TBD |

Table 4: Prevalence of six safety-judge failure modes in the TVD-Conflict dataset. Counts and shares will be inserted after annotation.

| Failure Mode | Count | Share (%) |
|---|---|---|
| Criteria ambiguity | TBD | TBD |
| Refusal but harmful | TBD | TBD |
| Compliance but safe | TBD | TBD |
| 50–50 problem | TBD | TBD |
| Distraction | TBD | TBD |
| Insufficient capability | TBD | TBD |

**Evaluation under a fixed search budget.** Table 3 will report Conflict Rate after a budget of ten candidate revisions, using the same conflict definitions as Table 2.

The resulting conflict set supports a systematic analysis of the recurring properties that lead jailbreak judges to different conclusions.

## 5 A Taxonomy of Jailbreak Judge Failure Modes

We analyze the conflict cases collected by TVD-Conflict and organize them into six recurring failure modes. These categories describe the source of the judgment error rather than the surface form of the jailbreak. Table 4 will report their prevalence after the conflict set has been annotated.

**Criteria ambiguity.** The judges operationalize the stated safety criterion differently, such as treating harmfulness, policy violation, and attack success as interchangeable.

**Refusal but harmful.** A response begins with refusal language but subsequently provides information that materially advances the harmful request. Some judges anchor on the refusal marker, while others evaluate the supplied content.

**Compliance but safe.** A response adopts the form of compliance without delivering harmful or usable information. Judges that rely on affirmative framing can mistake stylistic compliance for harmful assistance.

**The 50–50 problem.** The available evidence supports two plausible readings of comparable strength, causing verdicts to split across judges or rubric formulations.

**Distraction.** An evaluator focuses on wrappers, disclaimers, fictional settings, mathematical notation, or peripheral detail instead of the behavior that determines safety.

**Insufficient capability.** The judge fails to decode, follow, or reason over the representation required to identify the harmful behavior, including obfuscation and multi-step context.

## 6 Conclusion

Safety evaluation with LLM judges is not invariant to the evaluator. Across ten jailbreak attacks, three judge backbones and two rubrics produce materially different decisions on identical interactions. Cross-judge Conflict Rate is 24.2% under binary judging and 19.0% under scoring, with

substantially higher rates for particular attacks. The variation is strongly attack dependent, indicating that a single aggregate ASR can conceal systematic blind spots. TVD-Conflict turns these discrepancies into a data-generation objective and supports analysis through six operational failure modes. The resulting evidence argues for evaluating safety judges as fallible measurement instruments rather than treating their outputs as interchangeable ground truth.

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